

HOMEWORK 3
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), FALL, 2018

CHECK: 02/22/2019 DUE: 02/25/2019

SECTION 1.8 - INTRODUCTION TO LINEAR TRANSFORMATIONS

1. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Show that if T maps two linearly independent vectors in $\mathbb{R}^n$ onto a linearly dependent set of vectors $\{T(v_1), T(v_2)\}$ in $\mathbb{R}^m$, then the equation $T(x) = 0$ has a nontrivial solution.
2. Let $\{u, v\}$ be a set of linearly independent vectors in $\mathbb{R}^3$. The parametric equation of a plane P through u, v , and 0 is $x = su + tv$, where $s, t \in \mathbb{R}$, i.e.

$$P = \{su + tv \in \mathbb{R}^3 : s, t \in \mathbb{R}\}.$$

Suppose $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is some arbitrary linear transformation. What must be true about $T(u)$ and $T(v)$ in order for the image of the plane P to be a plane?

SECTION 1.9 - THE MATRIX OF A LINEAR TRANSFORMATION

3. Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $v_1 = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$, and $v_2 = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$, and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that maps x to $x_1 v_1 + x_2 v_2$, i.e.

$$T(x) = x_1 v_1 + x_2 v_2.$$

Find a matrix A such that $T(x) = Ax$ for all $x \in \mathbb{R}^2$.

4. Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation which maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ (i.e. T is *surjective*). What relationship between n and m must be true?
 5. Let $S : \mathbb{R}^p \rightarrow \mathbb{R}^n$ be a linear transformation, and let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Show that the mapping $F : \mathbb{R}^p \rightarrow \mathbb{R}^m$ given by $F(x) = T(S(x))$ is a linear transformation.
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SECTION 2.1 - MATRIX OPERATIONS

6. Let $A = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix}$, and $C = \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$. Verify that $AB = AC$ despite the fact that $B \neq C$.
7. For two vectors $u, v \in \mathbb{R}^n$, we say the 1×1 matrix (i.e. real number) $u^T v$ is the *inner product* (or *dot product*) of u and v , and we denote the inner product of u and v by $u \cdot v$ (i.e. $u \cdot v = u^T v$). Show that every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}$ can be written in the form $T(x) = v \cdot x$ for some $v \in \mathbb{R}^n$.
8. For two vectors $u, v \in \mathbb{R}^n$, we say the $n \times n$ matrix uv^T is the *outer product* of u and v , and we denote the outer product by $u \otimes v = uv^T$. Let $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $e_2 = 01$. Show that every 2×2 matrix M with real entries can be written in the form

$$M = c_1(e_1 \otimes e_1) + c_2(e_1 \otimes e_2) + c_3(e_2 \otimes e_1) + c_4(e_2 \otimes e_2)$$

for some constants $c_1, c_2, c_3, c_4 \in \mathbb{R}$.

(In general, a version of this formula holds for arbitrary $n \times n$ matrices. But we'll only do the 2×2 case here, since the general formula is more cumbersome to deal with.)